

Task 4: SQL & Database Management

Data Analyst Internship

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Business Query Results

Query 1: Total Sales by Product Line

	A	B
1	productline	totalsales
2	Classic Cars	3842868.54
3	Vintage Cars	1806675.68
4	Trucks and Buses	1111559.19
5	Motorcycles	1103512.19
6	Planes	969323.42
7	Ships	700039.22
8	Trains	226243.47
9		

Query 2: Average Order Value

Data Output		Messages	Notifications
			
			SQL
	averageordervalue double precision		
1	32752.421845637557		

Query 3: Customer Segmentation

Data Output		Messages	Notifications
			
			SQL
Showing rows: 1 to 89 			
	customername character varying (255)	totalspent double precision	customersegment text
1	Euro Shopping Channel	912294.1100000002	High Value
2	Mini Gifts Distributors Ltd.	654858.06	High Value
3	Australian Collectors, Co.	200995.40999999997	High Value
4	Muscle Machine Inc	197736.93999999997	High Value
5	La Rochelle Gifts	180124.9	High Value
6	Dragon Souvenirs, Ltd.	172989.68000000008	High Value
7	Land of Toys Inc.	164069.44000000003	High Value
8	The Sharp Gifts Warehouse	160010.26999999996	High Value
9	AV Stores. Co.	157807.80999999997	High Value
Total rows: 89		Query complete 00:00:00.267	

Query Optimization Analysis

1. Before Optimization (Sequential Scan)

	QUERY PLAN text
1	Seq Scan on auto_sales (cost=0.00..107.34 rows=1 width=167) (actual time=0.987..0.988 rows=0.00 loop...
2	Filter: ((customername)::text = 'Euro+ Shopping Channel'::text)
3	Rows Removed by Filter: 2747
4	Buffers: shared hit=73
5	Planning:
6	Buffers: shared hit=50 dirtied=2
7	Planning Time: 0.591 ms
8	Execution Time: 1.031 ms

The query was slow because it used a **Seq Scan**, forcing the database to read the entire table. The execution time was **0.589 ms**.

2. After Optimization (Index Scan)

	QUERY PLAN text
1	Index Scan using idx_customername on auto_sales (cost=0.28..8.30 rows=1 width=167) (actual time=0.155..0.155 rows=0.00 loo...
2	Index Cond: ((customername)::text = 'Euro+ Shopping Channel'::text)
3	Index Searches: 1
4	Buffers: shared read=2
5	Planning:
6	Buffers: shared hit=15 read=1
7	Planning Time: 6.578 ms
8	Execution Time: 0.192 ms

After creating an **INDEX** on the CUSTOMERNAME column, the query became much faster. It used an **Index Scan** to find the data instantly. The execution time dropped to **0.063 ms**.